# MCGI: Manifold-Consistent Graph Indexing for Billion-Scale Disk-Resident Vector Search

**Dongfang Zhao**
University of Washington, Tacoma School of Engineering and Technology
dzhao@uw.edu

## Abstract

Graph-based Approximate Nearest Neighbor (ANN) search often suffers from performance degradation in high-dimensional spaces due to the Euclidean-Geodesic mismatch, where greedy routing diverges from the underlying data manifold. To address this challenge, this paper presents Manifold-Consistent Graph Indexing (MCGI), a geometry-aware and disk-resident indexing method that leverages Local Intrinsic Dimensionality (LID) to dynamically adapt search strategies to the intrinsic geometry of data. Unlike conventional algorithms that treat dimensions uniformly, MCGI modulates its beam search budget based on in-situ geometric analysis, which reduces sensitivity to data-specific hyperparameters by replacing a single scalar with a geometry-informed range that remains stable across datasets of varying dimensionality. Theoretical analysis demonstrates that MCGI provides robust approximation by preserving manifold-consistent topological connectivity. Evaluations against three industry-standard baselines across four datasets up to billion scales confirm the advantages of the proposed approach.

## 1 Introduction

The advent of Large Language Models [4, 39] has fundamentally transformed the landscape of information retrieval and knowledge management. To address the inherent limitations of LLMs, such as hallucinations [19] and knowledge cutoff dates, Retrieval-Augmented Generation (RAG) [27] has emerged as a critical architectural paradigm. RAG relies heavily on the ability to retrieve semantically relevant context from massive corpora in real-time, typically via dense vector representations [24]. This dependency has placed Approximate Nearest Neighbor Search (ANNS) at the core of modern data infrastructure, demanding vector indices that can scale to billion-point datasets [35] while maintaining low latency and high recall under strict production constraints.

For extreme-scale embedding vector applications, many ANNS techniques have noticeably converged on graph-based indices, with DiskANN (Vamana) and its variants [36, 5, 11, 12] being representative solutions for large-scale, SSD-resident workloads. These algorithms typically employ greedy routing on a proximity graph to navigate from an entry point to the query target. While such methods exhibit exceptional performance on popular benchmarks like SIFT1M [21] (128 dimensions), their efficiency degrades significantly in high-dimensional spaces, such as GIST1M (960 dimensions). This degradation is often attributed to the curse of dimensionality [2], where the distance contrast diminishes, and the Euclidean shortest path on the graph diverges from the geodesic path on the underlying data manifold [37, 18]. We refer to this phenomenon as the *Euclidean-Geodesic Mismatch*, or EGM. When the routing algorithm ignores the intrinsic geometry of the data, it performs excessive backtracking and disk I/O, rendering the search inefficient, which is particularly problematic in high-recall scenarios where production systems often expect Recall@10 higher than 95%.

Our observation on this EGM problem is that high-dimensional real-world data is rarely uniformly distributed; Instead, it typically adheres to the Manifold Hypothesis [37, 33], residing on lower-

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

dimensional structures embedded within the ambient space. Consequently, the search difficulty is not uniform across the dataset but is modulated by the Local Intrinsic Dimensionality (LID) [26]. That is, in regions where the data manifold is flat (low LID), greedy routing is effective; however, in regions with high curvature or complex topology (high LID), standard greedy strategies may fail to identify the descent direction. *We thus argue that an optimal indexing strategy must be manifold-aware, dynamically allocating computational resources based on the local geometric complexity.*

To demonstrate the above hypothesis, we design Manifold-Consistent Graph Indexing (MCGI), a geometry-aware disk-based indexing method crafted to align Euclidean search with the underlying data manifold. By incorporating LID estimation directly into the routing logic, MCGI demonstrates the ability to adapt its traversal strategy to the local topology of the underlying manifold.

*The remainder of this paper will present the following contributions:* (1) A theoretical framework embedding the local intrinsic dimensionality to graph navigability with formal guarantees on estimation robustness, fixed-beam suboptimality, and construction-time adaptation; (2) Geometry-aware indexing algorithms that adapt pruning parameters without data-specific hyperparameters; and (3) $3\times$ lower latency than DiskANN at 90% recall on billion-scale vectors (SIFT1B), with consistent gains on billion-scale CLIP embeddings (T2I-1B).

## 2 Related Work

**Vector Indexing Paradigms** While traditional sparse retrieval methods rely on lexical matching [32], memory-resident graph indices like HNSW [29] establish state-of-the-art dense retrieval but face billion-scale memory bottlenecks. Disk-based approaches, such as DiskANN [36], SPANN [5], PipeANN [11], and OdinANN [12], along with predecessors like NSG [9], adapt topologies or inverted structures for SSDs. Recent advancements such as CSPG [41] further optimize sparse proximity graphs to reduce unpromising explorations, yet these methods predominantly rely on static routing parameters evaluated on moderate dimensionality. In contrast, MCGI dynamically modulates the search with local topology to align graph traversal with the underlying manifold.

**High-Dimensional Indexing** Early space-partitioning trees like the KD-tree [3] and R-tree [14] severely degrade to linear-scan performance in dimensions higher than 20 due to the curse of dimensionality. Approximate methods address this through sub-linear hashing like LSH [17, 7], which incurs excessive storage overhead, or subspace quantization like PQ [21] and OPQ [10], and randomized structures like RP-Trees [6], which sacrifice accuracy through compression. Furthermore, recent literature extensively studies the theoretical limits of navigable graphs [8] and the efficiency of index construction [43] in strictly high-dimensional scenarios. MCGI circumvents aggressive quantization losses and redundant hashing overheads by preserving the exact connectivity benefits of graph traversal adapted to local topology.

**Intrinsic Dimensionality** Theoretical metrics such as the doubling dimension [13] and expansion dimension [23] provide asymptotic bounds on search complexity, and recent analyses establish worst-case performance limits for popular implementations like DiskANN and HNSW based on intrinsic dimensionality [18]. Maximum likelihood estimators [26] for Local Intrinsic Dimensionality (LID) enable robust practical estimations. Existing works typically utilize LID passively for query hardness prediction [15, 40] or detecting adversarial examples [28] without altering the underlying index structure. Diverging from passive descriptive applications, MCGI actively utilizes LID as a prescriptive control signal to dynamically modulate graph traversal parameters for efficient routing.

## 3 Methodology

### 3.1 Mapping Function from LID to Pruning Parameter

We start by briefly reviewing the notions of Local Intrinsic Dimensionality (LID) in the words of analysis. A full treatment can be found, for example, in Houle [16].

**Definition 3.1** (Local Intrinsic Dimensionality)**.** Let $\mathcal{X}$ be a domain equipped with a distance measure $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}^+$. For a reference point $x \in \mathcal{X}$, let $F_x(r) = \mathbb{P}(d(x, Y) \leq r)$ denote the cumulative distribution function (CDF) of the distance between $x$ and a random variable $Y$ drawn from the

underlying data distribution, where $\mathbb{P}$ denotes the probability function and $r$ denotes the radius. The Local Intrinsic Dimensionality (LID) of $x$, denoted as $\mathrm{LID}(x)$, is defined as the intrinsic growth rate of the probability measure within the neighborhood of $x$:

$$\mathrm{LID}(x) \triangleq \lim_{r \to 0} \frac{r \cdot F'_x(r)}{F_x(r)} = \lim_{r \to 0} \frac{d \ln F_x(r)}{d \ln r}, \tag{1}$$

provided that the limit exists and $F_x(r)$ is continuously differentiable for $r > 0$.

While Definition 3.1 is elegant, it is often impractical to compute directly. According to [1], LID can be estimated as follows.

**Definition 3.2** (LID Maximum Likelihood Estimator)**.** Given a reference point $x$ and its $k$-nearest neighbors determined by the distance measure $d$. Let $r_i = d(x, v_i)$ denote the distance to the $i$-th nearest neighbor, sorted such that $r_1 \leq \cdots \leq r_k$. Following the formulation in [1], which adapts the Hill estimator for intrinsic dimensionality, the LID at $x$ can be estimated as:

$$\widehat{\mathrm{LID}}(x) = -\left( \frac{1}{k} \sum_{i=1}^{k} \ln \frac{r_i}{r_k} \right)^{-1}. \tag{2}$$

The primary hypothesis of the proposed Manifold-Consistent Graph Indexing (MCGI) is that the graph topology should adapt to the local geometric complexity, rather than being a static vertex-edge structure. To preserve topological fidelity, the indexing algorithm must adopt a conservative pruning strategy (smaller $\alpha$ in [36]), thereby forcing the search to take smaller, safer steps along the manifold surface. The question then becomes how to quantitatively map the estimated LID to the pruning parameter $\alpha$ that controls the edge selection during graph construction. We refer to such a function as the *mapping function* $\Phi : \mathbb{R}^+ \to [\alpha_{\min}, \alpha_{\max}]$, which translates the geometric complexity of the local neighborhood into a corresponding pruning constraint.

Formally, let $u \in V$ be a node in the graph and $\widehat{\mathrm{LID}}(u)$ be its estimated LID. We define the *pruning parameter* $\alpha(u)$ as:

$$\alpha(u) \triangleq \Phi(\widehat{\mathrm{LID}}(u)), \tag{3}$$

where $\Phi$ will be constructed to satisfy the following geometric requirements: in regions with high LID, the graph should enforce a stricter connectivity constraint (smaller $\alpha$) to avoid short-circuiting the manifold; conversely, in low-LID regions, the constraint can be relaxed (larger $\alpha$).

To ensure the mapping is robust across datasets with varying complexity scales, we employ Z-score normalization based on the empirical distribution of the LID estimates. We first compute the normalized score $z(u)$:

$$z(u) = \frac{\widehat{\mathrm{LID}}(u) - \mu_{\widehat{\mathrm{LID}}}}{\sigma_{\widehat{\mathrm{LID}}}}, \tag{4}$$

where $\mu_{\widehat{\mathrm{LID}}}$ and $\sigma_{\widehat{\mathrm{LID}}}$ denote the mean and standard deviation of the set of estimated LID values $\{\widehat{\mathrm{LID}}(v) \mid v \in V\}$ computed across the entire graph. We then formulate $\Phi$ using a logistic function to smoothly map the Z-score to the operational range $[\alpha_{\min}, \alpha_{\max}]$:

$$\Phi(\widehat{\mathrm{LID}}(u)) = \alpha_{\min} + \frac{\alpha_{\max} - \alpha_{\min}}{1 + \exp(z(u))}. \tag{5}$$

The mapping function $\Phi$ satisfies the following geometric properties that are essential for stable graph construction: Monotonicity and Boundedness. We state these properties in the following as propositions with proof sketches and provide full proofs in the Appendix.

**Proposition 3.3** (Monotonicity)**.** *The mapping function $\Phi$ is strictly decreasing with respect to the estimated local intrinsic dimensionality. Formally, given that the standard deviation of the LID estimates $\sigma_{\widehat{LID}} > 0$ and the pruning range $\alpha_{\max} > \alpha_{\min}$, the derivative satisfies:*

$$\frac{d\Phi}{d\widehat{LID}(u)} < 0. \tag{6}$$

*Proof Sketch.* By the chain rule, $\frac{d\Phi}{d\widehat{\text{LID}}(u)} = \frac{d\Phi}{dz} \cdot \frac{dz}{d\widehat{\text{LID}}(u)}$, where the Z-score satisfies $\frac{dz}{d\widehat{\text{LID}}(u)} = \frac{1}{\sigma_{\widehat{\text{LID}}}} > 0$. The derivative of the logistic component evaluates to $\frac{d\Phi}{dz} = -(\alpha_{\max} - \alpha_{\min}) \cdot \frac{e^z}{(1+e^z)^2}$, which is strictly negative since $\alpha_{\max} > \alpha_{\min}$ and $e^z > 0$ for all $z \in \mathbb{R}$. Combining these two terms yields a strictly negative derivative: $\alpha(u)$ decreases monotonically as geometric complexity increases. □

**Proposition 3.4** (Boundedness)**.** *The pruning parameter $\alpha(u)$ derived from the mapping function is strictly bounded within the prescribed operational interval. That is, for any node $u$ with a finite LID estimate, the following holds:*

$$\alpha_{\min} < \alpha(u) < \alpha_{\max}. \tag{7}$$

*Proof Sketch.* For any finite $\widehat{\text{LID}}(u)$, the Z-score $z(u)$ is finite, and the logistic function satisfies $S(u) = \frac{1}{1+e^{z(u)}} \in (0, 1)$ strictly, since $e^{z(u)} \in (0, \infty)$. Substituting into $\alpha(u) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot S(u)$ and applying the strict bounds on $S(u)$ directly yields $\alpha_{\min} < \alpha(u) < \alpha_{\max}$. □

### 3.2 Indexing Algorithms

The Manifold-Consistent Graph Indexing (MCGI) algorithm introduces a geometric calibration phase to a graph indexing procedure. Unlike static methods that apply a uniform connectivity rule, MCGI executes in two distinct stages to ensure that the topology respects the manifold structure, as follows. The full pseudo-code of the algorithm is in Appendix A.3, Algorithm 1.

**Phase 1: Geometric Calibration.** Before modifying the graph topology, the system first performs a global analysis of the dataset geometry. We estimate the LID for every point and aggregate the population statistics $(\mu, \sigma)$ defined in Section 3.1. This phase freezes the geometric profile of the dataset. By pre-computing these statistics, we decouple the complexity estimation from the graph update loop, ensuring that the mapping function $\Phi$ remains stable and computationally efficient during the intensive edge-selection process.

**Phase 2: Manifold-Consistent Refinement.** The index construction follows an iterative refinement strategy. Let $\mathcal{N}(u)$ denote the set of neighbors for node $u$ in the graph $G$. In each iteration, the algorithm dynamically updates $\mathcal{N}(u)$ by considering a candidate pool $\mathcal{C}$ obtained from a greedy search on the current graph structure:

1. Queries the pre-computed geometric profile to determine the node-specific constraint $\alpha(u)$.
2. Explores the graph to identify a candidate pool $\mathcal{C}$.
3. Filters connections using the dynamic occlusion criterion.

**Remark** The calibration phase takes $O(N \log N)$ dominated by the construction phase, on par with the complexity of DiskANN [36]. In practice, calculating the exact LID for the entire dataset may cause a computational bottleneck on billion-scale data. To address this, we also design an online algorithm called *Online-MCGI*, whose pseudo-code can be found in Appendix A.3, Algorithm 2.

### 3.3 Theoretical Analysis

#### 3.3.1 Geometric Complexity and Adaptive Routing

The efficiency of greedy routing on a proximity graph is intrinsically tied to the local topological properties of the underlying manifold $\mathcal{M}$. Under the Manifold Hypothesis, the search space is locally characterized by the Local Intrinsic Dimensionality (LID) at query $q$. We formalize the relationship between LID and the computational cost required to identify the nearest neighbor, as follows.

**Lemma 3.5** (Local Complexity Lower Bound)**.** *Consistent with the complexity bounds established for growth-restricted metrics [23], for a query $q$ on a manifold $\mathcal{M}$, the expected number of distance evaluations $N_{dist}$ required for successful greedy routing scales exponentially with the local intrinsic dimensionality $d = LID(q)$:*

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right), \tag{8}$$

*where $\lambda > 0$ is a geometric constant derived from the required convergence rate.*

*Proof Sketch.* At a node $u$ with distance $r$ to query $q$, a greedy step succeeds only if a neighbor $v$ falls within a cone of half-angle $\theta_{\max}$ around the direction of $q$; the probability of this event corresponds to the ratio of a spherical cap to the full hypersphere $\mathcal{S}^{d-1}$. Asymptotic analysis of this ratio via the regularized incomplete beta function yields $P_{\text{success}} \propto d^{-1/2}\exp(-\lambda \cdot d)$, where the exponential decay arises from the concentration of measure on high-dimensional spheres and $\lambda = -\ln(\sin\theta_{\max}) > 0$. Since $\mathbb{E}[N_{dist}] = 1/P_{\text{success}}$, the lower bound follows directly. □

Lemma 3.5 reveals a fundamental geometric barrier: for a fixed budget $L$, the probability of routing failure grows exponentially in high-LID regions. To address this, MCGI adopts an *Iso-Recall* strategy, aiming to homogenize the search reliability across the heterogeneous manifold.

**Proposition 3.6** (Optimal Budget Allocation)**.** *To maintain a uniform bound on the routing failure probability $\delta$ across the manifold (i.e., $\mathbb{P}(\textit{fail} \mid q) \le \delta$ for all $q \in \mathcal{M}$), the search beam width $L(q)$ must scale exponentially with the local intrinsic dimensionality:*

$$L(q) \propto \exp\left(\lambda \cdot \textit{LID}(q)\right). \tag{9}$$

*Proof Sketch.* For a beam of width $L$, the failure probability is $(1 - P_{\text{success}})^L \le \delta$, which via the approximation $\ln(1-x) \approx -x$ requires $L \ge -\ln\delta / P_{\text{success}}$. Substituting the bound $P_{\text{success}} \propto d^{-1/2}\exp(-\lambda \cdot d)$ from Lemma 3.5 and absorbing the polynomial and confidence factors into a constant $C_\delta$ yields $L(q) \ge C_\delta \cdot \exp(\lambda \cdot \text{LID}(q))$. □

While Proposition 3.6 establishes that the search budget $L$ must theoretically adapt to local geometric complexity, dynamically modulating $L$ per node during online serving disrupts memory alignment and degrades SIMD parallel efficiency. Therefore, in practical system deployments, $L$ is strictly bounded as a constant.

### 3.3.2 Topological Fidelity and Connectivity

A primary theoretical concern with aggressive edge pruning is the potential fracture of the connectivity backbone [42]. We will show that MCGI guarantees global reachability by strictly preserving the underlying manifold skeleton.

Formally, the edge selection in MCGI is governed by the adaptive pruning parameter $\alpha(u)$. Specifically, an edge $(u, v)$ is pruned if there exists a witness node $n$ such that $\alpha(u) \cdot d(n, v) \le d(u, v)$. This condition defines an *exclusion region* around the midpoint of $u$ and $v$. As $\alpha(u)$ increases, this region shrinks, making pruning more conservative. We formulate this in the following proposition.

**Proposition 3.7** (Connectivity Preservation)**.** *Let $G_{EMST}$ and $G_{RNG}$ denote the Euclidean Minimum Spanning Tree (EMST) and the Relative Neighborhood Graph (RNG), respectively. For any configuration of points in general position, provided that $\alpha(u) \ge 1.0$ for all $u \in V$, the graph $G_{MCGI}$ satisfies the strict inclusion hierarchy:*

$$E_{EMST} \subseteq E_{RNG} \subseteq E_{MCGI}. \tag{10}$$

*Consequently, $G_{MCGI}$ is connected.*

*Proof Sketch.* The first inclusion $E_{EMST} \subseteq E_{RNG}$ follows from the classical result of Toussaint [38], which holds for any finite point set in a metric space. For the second inclusion $E_{RNG} \subseteq E_{MCGI}$, note that MCGI prunes an edge $(u, v)$ only if a witness node falls within a exclusion region of radius $\frac{1}{\alpha(u)}d(u, v)$ around $v$; since $\alpha(u) \ge 1.0$, this exclusion region is a strict geometric subset of the RNG exclusion region, so any edge surviving the RNG pruning condition is guaranteed to survive MCGI pruning. Connectivity of $G_{MCGI}$ follows immediately from the inclusion hierarchy, since $G_{EMST}$ spans all vertices by definition. □

Proposition 3.7 ensures that there are no structural dead ends. While Lemma 3.5 dictates the *cost* of routing, Proposition 3.7 ensures the *viability* of routing. Even in the worst-case scenario where heuristics fail, a path exists from the entry point to any target via the edges of the underlying EMST, provided the graph traversal algorithm has sufficient width to discover these backbone links.

### 3.3.3 Robustness to LID Estimation Error

A practical concern with the LID-based mapping is whether estimation noise from the finite-sample MLE (Definition 3.2) propagates into the pruning parameter $\alpha(u)$ in a destabilizing manner. We address this concern in the following.

The MLE estimator in Definition 3.2 is known to exhibit asymptotic variance of order $\mathrm{LID}^2/k$ [26], yielding a pointwise estimation error bounded as:

$$\left|\widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)\right| = O\left(\frac{\mathrm{LID}(u)}{\sqrt{k}}\right). \tag{11}$$

We characterize how this error propagates through the mapping function $\Phi$.

**Proposition 3.8** (Estimation Robustness)**.** *Let $\widehat{\mathrm{LID}}(u)$ be the MLE estimate computed with $k$ nearest neighbors. The induced perturbation on the pruning parameter satisfies:*

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \mathrm{LID}(u)}{4\,\sigma_{\widehat{\mathrm{LID}}}\,\sqrt{k}}. \tag{12}$$

*Proof Sketch.* By a first-order Taylor expansion, $|\Delta\alpha(u)|$ is bounded by $\sup_L \left|\frac{d\Phi}{dL}\right| \cdot |\epsilon|$, where $\epsilon = \widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)$ is the estimation error. The gradient magnitude is bounded by $\frac{\alpha_{\max}-\alpha_{\min}}{4\,\sigma_{\widehat{\mathrm{LID}}}}$, since the logistic derivative $\frac{e^z}{(1+e^z)^2}$ achieves its global maximum of $\frac{1}{4}$ at $z = 0$, providing an intrinsic noise attenuation factor regardless of LID magnitude. Substituting the known asymptotic MLE error $|\epsilon| = O\left(\mathrm{LID}(u)/\sqrt{k}\right)$ [26] yields the bound directly, confirming that $|\Delta\alpha(u)|$ decays as $O(1/\sqrt{k})$ and even modest $k$ suffices for stable $\alpha$ assignments. □

Proposition 3.8 reveals two structural properties. First, the perturbation decays as $O(1/\sqrt{k})$, confirming that even a modest neighborhood size (e.g., $k = 20$) provides stable $\alpha$ assignments. Second, the Sigmoid mapping contributes a factor of at most $1/4$ (the global maximum of the logistic derivative), acting as an intrinsic noise buffer regardless of the LID magnitude. This provides a principled lower bound on the required $k$ during the geometric calibration phase.

### 3.3.4 Theoretical Justification of Construction-time Adaptation

While Proposition 3.6 establishes that an *ideal* search system should allocate beam width $L^*(q) \propto \exp(\lambda \cdot \mathrm{LID}(q))$ at query time, implementing this schedule online is impractical due to SIMD alignment and memory access constraints (§3.3.1). We now formalize the cost of abandoning this schedule and show that how MCGI's construction-time adaptation recovers the theoretical optimum.

**Definition 3.9** (Iso-Recall Beam Schedule)**.** For a target failure probability $\delta$, the oracle beam schedule is defined as:

$$L^*(q) = C_\delta \cdot \exp(\lambda \cdot \mathrm{LID}(q)), \tag{13}$$

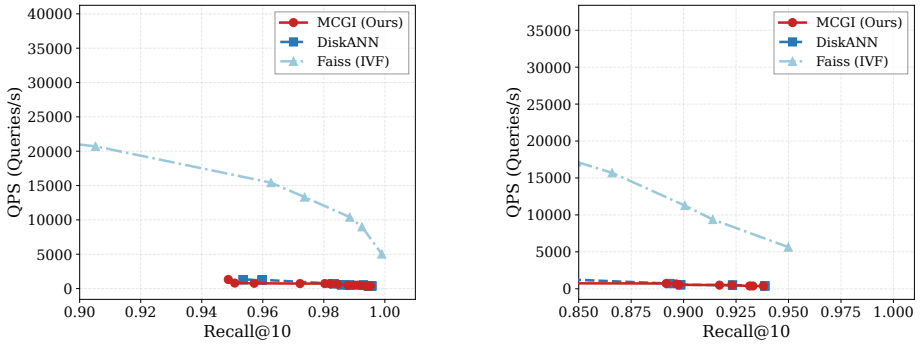
where $C_\delta = -\ln\delta / P_{\text{success}}$ is a recall-dependent constant. A search algorithm is said to be *recall-optimal* if it implements $L^*(q)$ for every query $q$.

**Proposition 3.10** (Suboptimality of Fixed Beam)**.** *For any fixed beam width $\bar{L}$, the recall gap in high-LID regions satisfies:*

$$P(\text{fail} \mid q) - \delta \;\geq\; 1 - \exp\left(-\frac{C_\delta}{\bar{L}}\exp\big(\lambda\big(\mathrm{LID}(q) - \mathrm{LID}_{\text{avg}}\big)\big)\right), \tag{14}$$

*which grows exponentially with the deviation of $\mathrm{LID}(q)$ from the dataset mean $\mathrm{LID}_{\text{avg}}$.*

*Proof Sketch.* For a fixed beam width $\bar{L}$, the failure probability is $(1 - P_{\text{success}})^{\bar{L}}$, while the oracle schedule $L^*(q) = C_\delta \exp(\lambda \cdot \mathrm{LID}(q))$ (Definition 3.9) achieves exactly $\delta$ by construction; the recall gap is thus the difference between these two quantities. In high-LID regions where $\mathrm{LID}(q) > \mathrm{LID}_{\text{avg}}$, the shortfall $\Delta L(q) = L^*(q) - \bar{L} > 0$ grows exponentially with $\mathrm{LID}(q)$, and applying the inequality $1 - e^{-x} \leq x$ to the resulting expression yields the closed-form lower bound in Eq. (14). The right-hand side is monotonically increasing in $\mathrm{LID}(q)$, so the recall deficit grows exponentially, creating a widening gap that cannot be compensated by any fixed-parameter strategy. □




(a) SIFT1M (128-dim)

(b) GloVe (100-dim)

Figure 1: Recall-QPS trade-off on million-scale datasets.

Proposition 3.10 quantifies precisely why static-parameter methods degrade on high-dimensional data: the recall deficit is not a constant offset but an *exponentially widening gap*. This provides a theoretical justification for geometry-aware adaptation that goes beyond empirical motivation. In practice, the following corollary establishes that MCGI's construction-time adaptation monotonically reduces this gap compared to any fixed-parameter baseline, structurally compensating for the shortfall of a constant beam width.

**Corollary 3.11** (Construction-time Structural Compensation for Fixed Beam Width)**.** *Under any fixed beam width $\bar{L}$, modulating $\alpha(u)$ during index construction according to $\Phi(\widehat{\mathrm{LID}}(u))$ monotonically reduces $P(\mathrm{fail} \mid q)$ for all queries in high-LID regions compared to any static-$\alpha$ baseline operating under the same $\bar{L}$. Specifically, nodes in high-LID regions receive conservative pruning (small $\alpha$), preserving a strictly larger navigational edge set that structurally compensates for the shortfall of $\bar{L}$ relative to the oracle schedule $L^*(q)$. This guarantee is monotone but not tight: it establishes that MCGI dominates fixed-$\alpha$ baselines, rather than quantifying the residual gap to the Iso-Recall Schedule $L^*(q)$ (Definition 3.9) directly.*

*Proof Sketch.* For nodes where $\mathrm{LID}(u) > \mathrm{LID}_{\mathrm{avg}}$, Proposition 3.3 guarantees $\alpha(u) < \bar{\alpha}$, which strictly shrinks the pruning exclusion region and thus preserves a superset of edges relative to any static-$\bar{\alpha}$ baseline, i.e., $E_{\mathrm{MCGI}} \supseteq E_{\bar{\alpha}}$ in high-LID regions. A strictly larger outgoing edge set monotonically increases $P_{\mathrm{success}}(q)$ at each beam step by the same geometric argument as Proposition 3.6, yielding $P(\mathrm{fail} \mid q, \mathrm{MCGI}) \leq P(\mathrm{fail} \mid q, \bar{\alpha})$ for all $q$ with $\mathrm{LID}(q) > \mathrm{LID}_{\mathrm{avg}}$. Note that this guarantee is monotone but not tight: quantifying the residual gap to the oracle schedule $L^*(q)$ (Definition 3.9) is left for future work. □

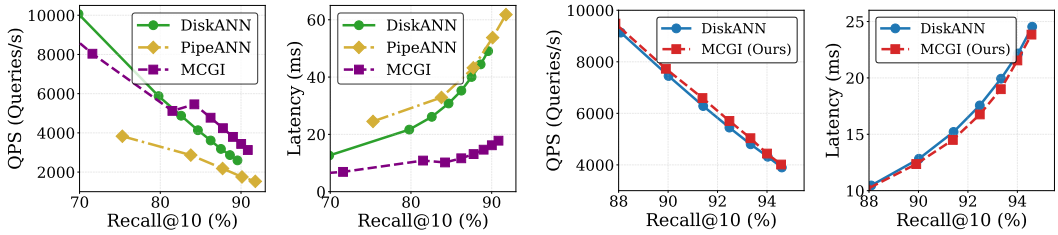
# 4 Evaluation

The anonymized code is at: `https://github.com/hpdic/mcgi`.

todo: need to update this with the new h100x4 testbed. All experiments were conducted with 96 CPU cores, 256 GiB of RAM, and 7.68 TB of NVMe SSD on Chameleon Cloud [25]. We evaluate MCGI on four benchmarks ranging from million-scale to billion-scale: SIFT1M [21], GloVe-100 [30], SIFT1B [22], and T2I-1B [34]. Our evaluation compares MCGI against three baselines: PipeANN [12, 11], DiskANN (Vamana) [36], and IVF-Flat (Faiss) [20]. More setup details are in Appendix A.2.

## 4.1 Million-scale Effectiveness

On the million-scale SIFT1M and GloVe-100 datasets, MCGI matches the throughput of DiskANN across the full recall range (Figures 1a and 1b). The in-memory Faiss IVF index serves as a throughput upper bound, since disk-resident indices pay SSD random-read latency on every routing hop. The parity with DiskANN shows that the adaptive pruning reduces to standard greedy search on these geometries without computational overhead.




(a) SIFT1B (b) T2I-1B

Figure 2: Throughput and latency of Online-MCGI on billion-scale datasets.

### 4.2 Billion-scale Scalability

We restrict billion-scale comparisons to PipeANN and DiskANN, which share the same Vamana backbone as MCGI and thus isolate the contribution of manifold-aware routing. All reported results on billion-scale datasets use Online-MCGI (Algorithm 2), where LID estimation is performed on-the-fly using the candidate set discovered during greedy search. The geometric calibration overhead is therefore fully included in the reported latency numbers. As shown in Figure 2a, at 90% recall MCGI maintains 3,436 QPS with 16.20 ms latency versus DiskANN's 2,597 QPS and 49.06 ms: a 3× latency reduction driven by fewer redundant disk accesses.

We further evaluated MCGI on the T2I-1B dataset in Figure 2b. The performance gap narrows compared to SIFT1B as expected: the I/O bottleneck is less pronounced when the compute cost per hop dominates. Nevertheless, MCGI maintains a consistent advantage at high-recall operating points (Recall@10 > 90%), confirming that manifold-aware pruning reduces redundant distance computations in CPU-bound scenarios.

## 5 Conclusion

This paper presents MCGI, a geometry-aware disk-resident indexing method for high-dimensional data of varying intrinsic dimensionality. MCGI resolves the Euclidean-Geodesic mismatch by adapting graph topology to local intrinsic dimensionality. Unlike existing methods that treat all regions of the underlying data manifold uniformly, MCGI dynamically allocates graph connectivity based on local geometric complexity, achieving what static-parameter approaches cannot: manifold-consistent routing that remains efficient as dimensionality scales and fluctuates. Our theoretical analysis formalizes this advantage with provable guarantees, and our empirical results confirm its effectiveness in practice: 3× lower latency than DiskANN at 90% recall on the billion-scale SIFT1B dataset, with consistent gains on billion-scale CLIP embeddings (T2I-1B).

## Acknowledgments and Disclosure of Funding

Results presented in this paper were obtained using the Chameleon testbed supported by the National Science Foundation. The system implementation was assisted by Copilot and Claude, which also helped polish the writing of this paper.

## References

[1] Laurent Amsaleg, Oussama Chelly, Teddy Furon, Stéphane Girard, Michael E. Houle, Ken-ichi Kawarabayashi, and Michael Nett. Estimating local intrinsic dimensionality. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '15, page 29–38, New York, NY, USA, 2015. Association for Computing Machinery.

[2] RICHARD BELLMAN and Stuart Dreyfus. *Dynamic Programming*, volume 33. Princeton University Press, 2010.

[3] Jon Louis Bentley. Multidimensional binary search trees used for associative searching. *Commun. ACM*, 18(9):509–517, September 1975.

[4] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20, Red Hook, NY, USA, 2020. Curran Associates Inc.

[5] Qi Chen, Bing Zhao, Haidong Wang, Mingqin Li, Chuanjie Liu, Zengzhong Li, Mao Yang, and Jingdong Wang. Spann: highly-efficient billion-scale approximate nearest neighbor search. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*, NIPS '21, Red Hook, NY, USA, 2021. Curran Associates Inc.

[6] Sanjoy Dasgupta and Yoav Freund. Random projection trees and low dimensional manifolds. In *Proceedings of the Fortieth Annual ACM Symposium on Theory of Computing*, STOC '08, page 537–546, New York, NY, USA, 2008. Association for Computing Machinery.

[7] Mayur Datar, Nicole Immorlica, Piotr Indyk, and Vahab S. Mirrokni. Locality-sensitive hashing scheme based on p-stable distributions. In *Proceedings of the Twentieth Annual Symposium on Computational Geometry*, SCG '04, page 253–262, New York, NY, USA, 2004. Association for Computing Machinery.

[8] Haya Diwan, Jinrui Gou, Cameron Musco, Christopher Musco, and Torsten Suel. Navigable graphs for high-dimensional nearest neighbor search: Constructions and limits. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 59513–59531. Curran Associates, Inc., 2024.

[9] Cong Fu, Chao Xiang, Changxu Wang, and Deng Cai. Fast approximate nearest neighbor search with the navigating spreading-out graph. *Proc. VLDB Endow.*, 12(5):461–474, January 2019.

[10] Tiezheng Ge, Kaiming He, Qifa Ke, and Jian Sun. Optimized product quantization. volume 36, page 744–755, USA, April 2014. IEEE Computer Society.

[11] Hao Guo and Youyou Lu. Achieving low-latency graph-based vector search via aligning best-first search algorithm with ssd. In *19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25)*, pages 171–186, Boston, MA, 2025. USENIX Association.

[12] Hao Guo and Youyou Lu. Odinann: Direct insert for consistently stable performance in billion-scale graph-based vector search. In *24th USENIX Conference on File and Storage Technologies (FAST 26)*, Santa Clara, CA, 2026. USENIX Association.

[13] A. Gupta, R. Krauthgamer, and J.R. Lee. Bounded geometries, fractals, and low-distortion embeddings. In *44th Annual IEEE Symposium on Foundations of Computer Science, 2003. Proceedings.*, pages 534–543, 2003.

[14] Antonin Guttman. R-trees: a dynamic index structure for spatial searching. In *Proceedings of the 1984 ACM SIGMOD International Conference on Management of Data*, SIGMOD '84, page 47–57, New York, NY, USA, 1984. Association for Computing Machinery.

[15] Junfeng He, Sanjiv Kumar, and Shih-Fu Chang. On the difficulty of nearest neighbor search. In *Proceedings of the 29th International Coference on International Conference on Machine Learning*, ICML'12, page 41–48, Madison, WI, USA, 2012. Omnipress.

[16] Michael E. Houle. Local intrinsic dimensionality I: an extreme-value-theoretic foundation for similarity applications. In Christian Beecks, Felix Borutta, Peer Kröger, and Thomas Seidl, editors, *Similarity Search and Applications - 10th International Conference, SISAP 2017, Munich, Germany, October 4-6, 2017, Proceedings*, volume 10609 of *Lecture Notes in Computer Science*, pages 64–79. Springer, 2017.

[17] Piotr Indyk and Rajeev Motwani. Approximate nearest neighbors: towards removing the curse of dimensionality. In *Proceedings of the Thirtieth Annual ACM Symposium on Theory of Computing*, STOC '98, page 604–613, New York, NY, USA, 1998. Association for Computing Machinery.

[18] Piotr Indyk and Haike Xu. Worst-case performance of popular approximate nearest neighbor search implementations: Guarantees and limitations. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 66239–66256. Curran Associates, Inc., 2023.

[19] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Comput. Surv.*, 55(12), March 2023.

[20] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with gpus. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.

[21] Herve Jégou, Matthijs Douze, and Cordelia Schmid. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1):117–128, 2011.

[22] Hervé Jégou, Romain Tavenard, Matthijs Douze, and Laurent Amsaleg. Searching in one billion vectors: Re-rank with source coding. In *2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 861–864, 2011.

[23] David R. Karger and Matthias Ruhl. Finding nearest neighbors in growth-restricted metrics. In *Proceedings of the Thiry-Fourth Annual ACM Symposium on Theory of Computing*, STOC '02, page 741–750, New York, NY, USA, 2002. Association for Computing Machinery.

[24] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online, November 2020. Association for Computational Linguistics.

[25] Kate Keahey, Jason Anderson, Zhuo Zhen, Pierre Riteau, Paul Ruth, Dan Stanzione, Mert Cevik, Jacob Colleran, Haryadi S. Gunawi, Cody Hammock, Joe Mambretti, Alexander Barnes, François Halbach, Alex Rocha, and Joe Stubbs. Lessons learned from the chameleon testbed. In *Proceedings of the 2020 USENIX Annual Technical Conference (USENIX ATC '20)*. USENIX Association, July 2020.

[26] Elizaveta Levina and Peter Bickel. Maximum likelihood estimation of intrinsic dimension. In L. Saul, Y. Weiss, and L. Bottou, editors, *Advances in Neural Information Processing Systems*, volume 17. MIT Press, 2004.

[27] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS '20, Red Hook, NY, USA, 2020. Curran Associates Inc.

[28] Xingjun Ma, Bo Li, Yisen Wang, Sarah M. Erfani, Sudanthi N. R. Wijewickrema, Michael E. Houle, Grant Schoenebeck, Dawn Song, and James Bailey. Characterizing adversarial subspaces using local intrinsic dimensionality. In *International Conference on Learning Representations*, 2018.

[29] Yu A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. volume 42, page 824–836, USA, April 2020. IEEE Computer Society.

[30] Jeffrey Pennington, Richard Socher, and Christopher D. Manning. Glove: Global vectors for word representation. In Alessandro Moschitti, Bo Pang, and Walter Daelemans, editors, *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing, EMNLP 2014, October 25-29, 2014, Doha, Qatar, A meeting of SIGDAT, a Special Interest Group of the ACL*, pages 1532–1543. ACL, 2014.

[31] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021.

[32] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. In *Foundations and Trends in Information Retrieval*, volume 3, pages 333–389. Now Publishers, Inc., 2009.

[33] Sam T. Roweis and Lawrence K. Saul. Nonlinear dimensionality reduction by locally linear embedding. *Science*, 290(5500):2323–2326, 2000.

[34] Harsha Vardhan simhadri, Martin Aumüller, Matthijs Douze, Dmitry Baranchuk, Amir Ingber, Edo Liberty, George Williams, Ben Landrum, Magdalen Dobson Manohar, Mazin Karjikar, Laxman Dhulipala, Meng Chen, Yue Chen, Rui Ma, Kai Zhang, Yuzheng Cai, Jiayang Shi, Weiguo Zheng, Yizhuo Chen, Jie Yin, and Ben Huang. Results of the big ANN: NeurIPS'23 competition. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2025.

[35] Harsha Vardhan Simhadri, George Williams, Martin Aumüller, Matthijs Douze, Artem Babenko, Dmitry Baranchuk, Qi Chen, Lucas Hosseini, Ravishankar Krishnaswamy, Gopal Srinivasa, Suhas Jayaram Subramanya, and Jingdong Wang. Results of the neurips'21 challenge on billion-scale approximate nearest neighbor search. In *NeurIPS 2021 Competitions and Demonstrations Track*, pages 177–189. PMLR, 2022.

[36] Suhas Jayaram Subramanya, Devvrit, Rohan Kadekodi, Ravishankar Krishaswamy, and Harsha Vardhan Simhadri. Diskann: fast accurate billion-point nearest neighbor search on a single node. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, 2019.

[37] Joshua B. Tenenbaum, Vin de Silva, and John C. Langford. A global geometric framework for nonlinear dimensionality reduction. *Science*, 290(5500):2319–2323, 2000.

[38] Godfried T. Toussaint. The relative neighbourhood graph of a finite planar set. *Pattern Recognition*, 12(4):261–268, 1980.

[39] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models. 2023.

[40] Zeyu Wang, Qitong Wang, Xiaoxing Cheng, Peng Wang, Themis Palpanas, and Wei Wang. Steiner-hardness: A query hardness measure for graph-based ann indexes. *Proc. VLDB Endow.*, 17(13):4668–4682, September 2024.

[41] Ming Yang, Yuzheng Cai, and Weiguo Zheng. Cspg: Crossing sparse proximity graphs for approximate nearest neighbor search. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 103076–103100. Curran Associates, Inc., 2024.

[42] Dongfang Zhao and Li Yang. Incremental isometric embedding of high-dimensional data using connected neighborhood graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 31(1):86–98, 2009.

[43] Xi Zhao, Yao Tian, Kai Huang, Bolong Zheng, and Xiaofang Zhou. Towards efficient index construction and approximate nearest neighbor search in high-dimensional spaces. *Proc. VLDB Endow.*, 16(8):1979–1991, April 2023.

## NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The paper presents both theoretical and experimental evidence to support its claims.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: We discuss the limitations of our approach by mentioning that the lower and upper bounds of the $\alpha$ cannot be left empty.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Full proofs are provided in the appendix, and all assumptions are clearly stated in the main paper and/or the appendix.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: All code and data needed to reproduce the main experimental results are availabe on the anonymous Github repository.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
     (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All the details to reproduce the main experimental results are provided in the anonymous Github repository, including instructions for data access and preparation, and scripts to reproduce all experimental results for the new proposed method and baselines.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: All such details are provided in the anonymous Github repository, and the main paper includes a summary of the experimental setting.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The results are pretty stable so we did not report error bars; we explain this in the main paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: These details are provided in the anonymous Github repository, and the main paper includes a summary of the compute resources used for the experiments.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: We believe we have followed the NeurIPS Code of Ethics in all aspects of our research.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: [N/A]

    Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: [N/A]

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [N/A]

    Justification: [N/A]

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [N/A]

    Justification: [N/A]

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: [N/A]

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: [N/A]

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.

- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: [N/A]

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

# A Technical appendices and supplementary material

## A.1 Complete Proofs

**Proposition A.1** (Monotonicity)**.** *The mapping function $\Phi$ is strictly decreasing with respect to the estimated local intrinsic dimensionality. Formally, given that the standard deviation of the LID estimates $\sigma_{\widehat{LID}} > 0$ and the pruning range $\alpha_{\max} > \alpha_{\min}$, the derivative satisfies:*

$$\frac{d\Phi}{d\widehat{LID}(u)} < 0. \tag{15}$$

*Proof.* Let $L = \widehat{\mathrm{LID}}(u)$ be the independent variable. We define the normalized Z-score $z$ as a function of $L$:

$$z(L) = \frac{L - \mu_{\widehat{\mathrm{LID}}}}{\sigma_{\widehat{\mathrm{LID}}}}. \tag{16}$$

The mapping function is defined as:

$$\Phi(L) = \alpha_{\min} + \frac{C}{1 + \exp(z(L))}, \tag{17}$$

where $C = \alpha_{\max} - \alpha_{\min}$. Since we strictly require $\alpha_{\max} > \alpha_{\min}$, it follows that $C > 0$. To determine the sign of the gradient, we apply the chain rule:

$$\frac{d\Phi}{dL} = \frac{d\Phi}{dz} \cdot \frac{dz}{dL}. \tag{18}$$

First, we differentiate the Z-score term with respect to $L$:

$$\frac{dz}{dL} = \frac{1}{\sigma_{\widehat{\mathrm{LID}}}}. \tag{19}$$

Next, we differentiate the logistic component $\Phi$ with respect to $z$:

$$\frac{d\Phi}{dz} = \frac{d}{dz}\left(\alpha_{\min} + C(1 + e^z)^{-1}\right) \tag{20}$$

$$= C \cdot (-1) \cdot (1 + e^z)^{-2} \cdot \frac{d}{dz}(1 + e^z) \tag{21}$$

$$= -C \cdot \frac{e^z}{(1 + e^z)^2}. \tag{22}$$

Combining these terms yields the full derivative:

$$\frac{d\Phi}{dL} = -\frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \cdot \frac{e^z}{(1 + e^z)^2}. \tag{23}$$

We analyze the sign of each component:

- The operational range constant $C > 0$.
- The standard deviation $\sigma_{\widehat{\mathrm{LID}}} > 0$, assuming the dataset exhibits non-zero geometric variance.
- The exponential function $e^z > 0$ for all $z \in \mathbb{R}$.
- The denominator $(1 + e^z)^2 > 0$.

Therefore, the term $\frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \frac{e^z}{(1+e^z)^2}$ is strictly positive. The leading negative sign guarantees that $\frac{d\Phi}{dL} < 0$. This confirms that the pruning parameter $\alpha$ strictly decreases as the local geometric complexity increases, thereby enforcing a more conservative graph topology in high-LID regions to prevent topology distortion in complex regions. □

**Proposition A.2** (Boundedness)**.** *The pruning parameter $\alpha(u)$ derived from the mapping function is strictly bounded within the prescribed operational interval. For any node $u$ with a finite LID estimate:*

$$\alpha_{\min} < \alpha(u) < \alpha_{\max}. \tag{24}$$

*Proof.* Let $S(u)$ denote the logistic component of the mapping function:

$$S(u) = \frac{1}{1 + \exp(z(u))}. \tag{25}$$

For any finite input $\widehat{\mathrm{LID}}(u)$, the Z-score $z(u)$ is finite. The exponential function maps the real line to the positive real line, i.e., $\exp(z(u)) \in (0, \infty)$. Consequently, the denominator lies in the interval $(1, \infty)$. Taking the reciprocal yields the bounds for the logistic component:

$$0 < S(u) < 1. \tag{26}$$

Substituting $S(u)$ back into the definition of $\Phi$:

$$\alpha(u) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot S(u). \tag{27}$$

Since $(\alpha_{\max} - \alpha_{\min}) > 0$, we can apply the inequality boundaries:

$$\alpha(u) > \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot 0 = \alpha_{\min}, \tag{28}$$

$$\alpha(u) < \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot 1 = \alpha_{\max}. \tag{29}$$

This proves that the topology is strictly confined. The pruning behavior never exceeds the relaxation upper limit ($\alpha_{\max}$) and never becomes stricter than the lower limit ($\alpha_{\min}$), ensuring graph connectivity and preventing degree explosion. □

**Lemma A.3** (Local Complexity Lower Bound)**.** *Consistent with the complexity bounds established for growth-restricted metrics [23], for a query $q$ on a manifold $\mathcal{M}$, the expected number of distance evaluations $N_{dist}$ required for successful greedy routing scales exponentially with the local intrinsic dimensionality $d = LID(q)$:*

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right), \tag{30}$$

*where $\lambda > 0$ is a geometric constant derived from the required convergence rate.*

*Proof.* Let the local dataset around a query $q$ be modeled as a uniform distribution on a manifold of intrinsic dimension $d$. Consider a greedy routing step where the algorithm is currently at a node $u$ with distance $r = \|u - q\|$.

**1. Geometric Condition for Improvement.** To make progress, the algorithm must identify a neighbor $v$ such that the distance to the query is reduced by a factor $\epsilon$ (where $0 < \epsilon < 1$), i.e., $\|v - q\| \leq (1 - \epsilon)r$. Let $s = \|u - v\|$ be the distance between the current node and the candidate neighbor (the step size). Applying the Law of Cosines to triangle $\triangle uvq$, we have:

$$\|v - q\|^2 = r^2 + s^2 - 2rs\cos\theta, \tag{31}$$

where $\theta$ is the angle between the vector $\vec{uq}$ and $\vec{uv}$. The improvement condition $\|v-q\|^2 \leq (1-\epsilon)^2 r^2$ implies:

$$\begin{aligned} r^2 + s^2 - 2rs\cos\theta &\leq (1-\epsilon)^2 r^2, \\ 2rs\cos\theta &\geq r^2 + s^2 - r^2(1-\epsilon)^2, \\ \cos\theta &\geq \frac{r^2 + s^2 - r^2(1-\epsilon)^2}{2rs}. \end{aligned} \tag{32}$$

Assuming the step size is small relative to the distance (i.e., $s \ll r$), the right-hand side is dominated by $1 - (1-\epsilon)^2 \approx 2\epsilon$, yielding a critical angle threshold $\theta_{max}$ such that any successful neighbor must fall within the cone defined by $0 \leq \theta \leq \theta_{max}$.

**2. Volume Concentration of the Improving Region.** In the tangent space at $u$, the directions of neighbors are uniformly distributed on the unit hypersphere $\mathbb{S}^{d-1}$. The probability $P_{success}$ of finding a neighbor in the improving cone corresponds to the ratio of the surface area of the spherical cap $\mathcal{C}(\theta_{max})$ to the total surface area of $\mathbb{S}^{d-1}$. This ratio is given by the regularized incomplete beta function, which can be expressed in terms of the integral over the polar angle $\phi$:

$$P_{success} = \frac{\text{Area}(\mathcal{C}(\theta_{max}))}{\text{Area}(\mathbb{S}^{d-1})} = \frac{\int_0^{\theta_{max}} \sin^{d-2}\phi \, d\phi}{\int_0^{\pi} \sin^{d-2}\phi \, d\phi}. \tag{33}$$

**3. Asymptotic Analysis.** We analyze the asymptotic behavior of this ratio as $d \to \infty$. The denominator relates to the surface area of the hypersphere and can be evaluated using the standard integral identity for powers of sine. Exploiting the symmetry of $\sin\phi$ around $\pi/2$, we have:

$$\begin{aligned}\int_0^{\pi} \sin^{d-2}\phi \, d\phi &= 2\int_0^{\frac{\pi}{2}} \sin^{d-2}\phi \, d\phi \\ &= 2 \cdot \frac{\sqrt{\pi}}{2} \frac{\Gamma(\frac{d-1}{2})}{\Gamma(\frac{d}{2})} = \sqrt{\pi}\frac{\Gamma(\frac{d-1}{2})}{\Gamma(\frac{d}{2})}.\end{aligned} \tag{34}$$

For high-dimensional spaces ($d \gg 1$), we apply the asymptotic property of the Gamma function ratio, $\frac{\Gamma(x+a)}{\Gamma(x)} \approx x^a$. Setting $x = d/2$ and $a = -1/2$, we obtain the approximation:

$$\sqrt{\pi}\frac{\Gamma(\frac{d}{2} - \frac{1}{2})}{\Gamma(\frac{d}{2})} \approx \sqrt{\pi}\left(\frac{d}{2}\right)^{-\frac{1}{2}} = \sqrt{\frac{2\pi}{d}}. \tag{35}$$

For the numerator, we exploit the concentration of measure near the integration boundary $\theta_{max}$. Since the integrand $\sin^{d-2}\phi$ decays exponentially fast away from $\theta_{max}$, the integral is dominated by the contribution near the upper limit. We apply integration by parts to extract the leading order term. Recall that $\frac{d}{d\phi}(\sin^{d-1}\phi) = (d-1)\sin^{d-2}\phi\cos\phi$. We rewrite the integrand as:

$$\int_0^{\theta_{max}} \sin^{d-2}\phi \, d\phi = \int_0^{\theta_{max}} \frac{1}{(d-1)\cos\phi}\frac{d}{d\phi}(\sin^{d-1}\phi)\, d\phi. \tag{36}$$

Neglecting lower-order terms generated by the derivative of $\frac{1}{\cos\phi}$, the integral approximates to the boundary term:

$$\left[\frac{\sin^{d-1}\phi}{(d-1)\cos\phi}\right]_0^{\theta_{max}} = \frac{\sin^{d-1}\theta_{max}}{(d-1)\cos\theta_{max}}. \tag{37}$$

Substituting these approximations back into the probability ratio, we obtain:

$$P_{success} \approx \frac{1}{\sqrt{2\pi}\cos\theta_{max}} \cdot \frac{\sqrt{d}}{d-1} \cdot (\sin\theta_{max})^{d-1}. \tag{38}$$

For large $d$, the term $\frac{\sqrt{d}}{d-1}$ is asymptotically equivalent to $d^{-1/2}$. Since $\theta_{max}$ is fixed by the geometric constraints, the trigonometric factors are constant relative to $d$. Let $\gamma = \sin\theta_{max} < 1$. The probability scales as:

$$P_{success} \propto d^{-1/2} \cdot \gamma^d = d^{-1/2}\exp(d\ln\gamma). \tag{39}$$

Defining the decay rate $\lambda = -\ln\gamma > 0$ (since $\gamma < 1$), we strictly obtain:

$$P_{success} \propto d^{-1/2}\exp(-\lambda \cdot d). \tag{40}$$

**4. Complexity Bound.** The expected number of distance evaluations follows $\mathbb{E}[N_{dist}] = 1/P_{success}$. Thus:

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\sqrt{d} \cdot \exp(\lambda \cdot d)\right). \tag{41}$$

This confirms that the routing cost is dominated by the exponential term $\exp(\lambda \cdot \mathrm{LID}(q))$, necessitating the adaptive beam width design in MCGI. □

**Proposition A.4** (Optimal Budget Allocation)**.** *To maintain a uniform bound on the routing failure probability $\delta$ across the manifold (i.e., $\mathbb{P}(\textit{fail} \mid q) \leq \delta$ for all $q \in \mathcal{M}$), the search beam width $L(q)$ must scale exponentially with the local intrinsic dimensionality:*

$$L(q) \propto \exp\left(\lambda \cdot \textit{LID}(q)\right). \tag{42}$$

*Proof.* Consider a beam search with width $L$. The probability of failing to find a descent step in one round is approximately $(1 - P_{success})^L$. To guarantee a recall target, we enforce a constant upper bound $\delta$ on the failure probability:

$$(1 - P_{success})^L \leq \delta. \tag{43}$$

Taking the natural logarithm and using the approximation $\ln(1-x) \approx -x$ for small $P_{success}$, we require:

$$-L \cdot P_{success} \leq \ln \delta \implies L \geq \frac{-\ln \delta}{P_{success}}. \tag{44}$$

Substituting the relationship $P_{success} = 1/\mathbb{E}[N_{dist}]$ established in Lemma 3.5, we obtain the asymptotic upper bound $P_{success} \leq \mathcal{O}(\sqrt{d} \cdot \exp(-\lambda \cdot d))$, where $d = \text{LID}(q)$. To satisfy the recall constraint in the worst case scenario, the search budget $L$ must grow inversely to $P_{success}$:

$$L \geq \frac{-\ln \delta}{\mathcal{O}(\sqrt{d} \cdot \exp(-\lambda \cdot d))} = \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right). \tag{45}$$

In high-dimensional spaces ($d \gg 1$), the exponential growth term strictly dominates the polynomial decay factor $1/\sqrt{d}$. By absorbing the lower-order polynomial variations and the confidence factor $-\ln \delta$ into a generalized constant $C_\delta$, the necessary budget asymptotically simplifies to:

$$L(q) \geq C_\delta \cdot \exp(\lambda \cdot \text{LID}(q)). \tag{46}$$

□

**Proposition A.5** (Connectivity Preservation)**.** *Let $G_{EMST}$ and $G_{RNG}$ denote the Euclidean Minimum Spanning Tree and the Relative Neighborhood Graph, respectively. For any configuration of points in general position, provided that $\alpha(u) \geq 1.0$ for all $u \in V$, the graph $G_{MCGI}$ satisfies the strict inclusion hierarchy:*

$$E_{EMST} \subseteq E_{RNG} \subseteq E_{MCGI}. \tag{47}$$

*Consequently, $G_{MCGI}$ is connected.*

*Proof.* The proof proceeds in two steps. First, we invoke the classical result established by Toussaint [38], which proves that the Relative Neighborhood Graph (RNG) is a supergraph of the Euclidean Minimum Spanning Tree (EMST) for any finite set of points in a metric space:

$$E_{EMST} \subseteq E_{RNG}. \tag{48}$$

Since the EMST connects all vertices in a single component, $G_{RNG}$ is necessarily connected.

Second, we show that $E_{RNG} \subseteq E_{MCGI}$. In the RNG, an edge between $u$ and $v$ is pruned if and only if a witness node falls within the intersection of two spheres centered at $u$ and $v$ with radius $d(u, v)$. We denote this geometric intersection as the exclusion region $\mathcal{R}_{RNG}$. Therefore, an edge exists in the RNG if and only if $\mathcal{R}_{RNG}$ contains no data points. In MCGI, the pruning condition requires the witness node to be within a distance of $\frac{1}{\alpha(u)} d(u, v)$ from $v$. Because the algorithm enforces $\alpha(u) \geq 1.0$, the pruning radius in MCGI is strictly less than or equal to the pruning radius of the RNG. Consequently, the exclusion region of MCGI is a geometric subset of the RNG exclusion region:

$$\mathcal{R}_{MCGI} \subseteq \mathcal{R}_{RNG}. \tag{49}$$

This subset relationship of exclusion regions creates an inverse superset relationship for the preserved edges. If an edge survives the RNG condition, its $\mathcal{R}_{RNG}$ is empty. Since $\mathcal{R}_{MCGI}$ is contained entirely within $\mathcal{R}_{RNG}$, it must also be empty. Thus, any edge present in the RNG is guaranteed to be preserved in MCGI, yielding the inclusion hierarchy of edge sets:

$$E_{RNG} \subseteq E_{MCGI}. \tag{50}$$

By combining this with the classical result that $E_{EMST} \subseteq E_{RNG}$ derived in [38], we establish the complete global connectivity chain. □

**Proposition A.6** (Estimation Robustness)**.** *Let $\widehat{\text{LID}}(u)$ be the MLE estimate computed with $k$ nearest neighbors. The induced perturbation on the pruning parameter satisfies:*

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \text{LID}(u)}{4\, \sigma_{\widehat{\text{LID}}} \sqrt{k}}. \tag{51}$$

*Proof.* Let $\epsilon = \widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)$ denote the estimation error. By the first-order Taylor expansion of $\Phi$ around $\mathrm{LID}(u)$:

$$|\Delta\alpha(u)| = \left|\Phi\Big(\widehat{\mathrm{LID}}(u)\Big) - \Phi(\mathrm{LID}(u))\right| \leq \sup_L \left|\frac{d\Phi}{dL}\right| \cdot |\epsilon|. \tag{52}$$

From the derivative computed in Proof A.1, the magnitude of the gradient is:

$$\left|\frac{d\Phi}{dL}\right| = \frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \cdot \frac{e^z}{(1+e^z)^2}, \tag{53}$$

where $C = \alpha_{\max} - \alpha_{\min}$. The logistic factor $\frac{e^z}{(1+e^z)^2}$ achieves its global maximum of $1/4$ at $z = 0$, so:

$$\left|\frac{d\Phi}{dL}\right| \leq \frac{\alpha_{\max} - \alpha_{\min}}{4\,\sigma_{\widehat{\mathrm{LID}}}}. \tag{54}$$

Substituting the asymptotic error bound from Eq. (11), namely $|\epsilon| = O(\mathrm{LID}(u)/\sqrt{k})$ [26], yields:

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \mathrm{LID}(u)}{4\,\sigma_{\widehat{\mathrm{LID}}}\,\sqrt{k}}. \tag{55}$$

This completes the proof. □

**Proposition A.7** (Suboptimality of Fixed Beam)**.** *For any fixed beam width $\bar{L}$, the recall gap in high-LID regions satisfies Eq.* (14)*, which grows exponentially with the deviation of* $\mathrm{LID}(q)$ *from the dataset mean.*

*Proof.* Consider a beam search with fixed width $\bar{L}$. From Proof A.4, the failure probability at query $q$ is:

$$P(\text{fail} \mid q) = \left(1 - P_{\text{success}}(q)\right)^{\bar{L}}. \tag{56}$$

From Lemma 3.5, we have:

$$P_{\text{success}}(q) \propto d^{-1/2} \exp(-\lambda \cdot d), \quad d = \mathrm{LID}(q). \tag{57}$$

For the oracle schedule $L^*(q)$ in Definition 3.9, the constraint $P(\text{fail} \mid q) \leq \delta$ holds by construction. For fixed $\bar{L}$, define $\Delta L(q) = L^*(q) - \bar{L}$. In high-LID regions where $\mathrm{LID}(q) > \mathrm{LID}_{\text{avg}}$, we have $\Delta L(q) > 0$ and:

$$\begin{aligned} P(\text{fail} \mid q) - \delta &= (1 - P_{\text{success}})^{\bar{L}} - (1 - P_{\text{success}})^{L^*(q)} \\ &\geq 1 - \exp\left(-\frac{C_\delta}{\bar{L}} \exp\big(\lambda\big(\mathrm{LID}(q) - \mathrm{LID}_{\text{avg}}\big)\big)\right), \end{aligned} \tag{58}$$

where the inequality applies $1 - e^{-x} \leq x$ and substitutes $L^*(q)$ from Definition 3.9. The right-hand side is monotonically increasing in $\mathrm{LID}(q)$, confirming the exponential widening of the recall gap. □

**Corollary A.8** (Construction-time Structural Compensation)**.** *Under any fixed beam width $\bar{L}$, the adaptive pruning in MCGI monotonically reduces $P(\text{fail} \mid q)$ for queries in high-LID regions compared to any static-$\alpha$ baseline operating under the same $\bar{L}$.*

*Proof.* We proceed in two steps: first showing that MCGI preserves a strictly larger edge set in high-LID regions, then connecting this to a lower failure probability under fixed beam search.

**Step 1: Edge set inclusion.** The pruning criterion in MCGI discards an edge $(u, v)$ if and only if there exists a witness node $n$ such that $\alpha(u) \cdot d(n, v) \leq d(u, v)$. For nodes where $\mathrm{LID}(u) > \mathrm{LID}_{\text{avg}}$, Proposition A.1 guarantees $\alpha(u) < \bar{\alpha}$. Since $\alpha(u) < \bar{\alpha}$ strictly shrinks the left-hand side of the pruning condition, any edge that would be pruned under $\alpha(u)$ is also pruned under $\bar{\alpha}$, but not vice versa. Therefore $\mathcal{E}_{\text{MCGI}} \supseteq \mathcal{E}_{\bar{\alpha}}$ in high-LID regions, with strict inclusion whenever at least one edge survives $\alpha(u)$ but not $\bar{\alpha}$.

**Step 2: Failure probability.** Under beam search with fixed width $\bar{L}$, the failure probability at query $q$ is governed by $P_{\text{success}}(q)$, the probability of finding a descent step from any node on the routing path (Proof A.4). A strictly larger outgoing edge set at each high-LID node monotonically increases

the number of candidate directions evaluated per beam step, which by the same geometric argument in Proof A.4 yields a weakly higher $P_{\text{success}}(q)$ and thus:

$$P(\text{fail} \mid q, \text{MCGI}) \leq P(\text{fail} \mid q, \bar{\alpha}), \quad \forall\, q \text{ s.t. } \text{LID}(q) > \text{LID}_{\text{avg}}, \tag{59}$$

with equality only when $\alpha(u) = \bar{\alpha}$ for all $u$ on the routing path, i.e., when the data is geometrically uniform. This establishes a monotone improvement over fixed-$\alpha$ baselines; quantifying the residual gap to the Iso-Recall Schedule $L^*(q)$ (Definition 3.9) is left for future work. □

### A.2 Additional Experimental Details

**Experiment Platform** All experiments were conducted on the Chameleon Cloud [25] platform using a compute node equipped with 96 AMD EPYC 7352 CPU cores and 256 GiB of RAM. The operating system is Ubuntu 24.04 LTS. All algorithms were compiled using GCC 11.4 with -O3 and AVX2 optimizations enabled to fully utilize the hardware instruction set. We utilize a single enterprise-grade Samsung PM1733 NVMe SSD (model MZWLJ7T6HALA) with a total physical capacity of 7.68 TB. In this configuration, the physical drive is divided into four independent hardware namespaces, each providing approximately 1.7 TiB of storage capacity. These four namespaces are subsequently aggregated into a single 7 TB logical volume using the Linux Logical Volume Manager (LVM). This setup leverages the PCIe Gen4 interface to deliver the massive I/O throughput and low latency.

**Datasets** We evaluate our method on four standard benchmarks, ranging from million-scale to billion-scale, to comprehensively test robustness across varying intrinsic dimensionalities and data volumes. In the following descriptions, $N$ denotes the number of base vectors and $D$ represents the vector dimensionality:

- SIFT1M [21] ($N = 10^6, D = 128$): A standard computer vision dataset using Euclidean distance ($L_2$).
- GloVe-100 [30] ($N = 1.2 \times 10^6, D = 100$): Word embedding vectors measuring semantic similarity. Following standard practice, we normalize the vectors to unit length and use Euclidean distance as a proxy for Cosine similarity.
- SIFT1B [22] ($N = 10^9, D = 128$): A billion-scale dataset used to evaluate the scalability and I/O efficiency of our method. It shares the same feature distribution as SIFT1M but scales the volume by three orders of magnitude, representing a realistic industrial scenario.
- T2I-1B [34] ($N = 10^9, D = 200$): A billion-scale dataset consisting of CLIP [31] embeddings derived from the LAION-5B dataset. Unlike visual descriptors like SIFT, T2I-1B represents a cross-modal retrieval scenario with a more complex manifold structure and higher intrinsic dimensionality, posing greater challenges for graph-based indexing algorithms.

**Baselines** We compare MCGI against three baselines, each serving a distinct role:

- PipeANN [11, 12]: A recent state-of-the-art disk-based system that optimizes search throughput via aggressive I/O pipeline parallelism. We include it to evaluate whether system-level asynchronous I/O optimizations can compensate for topological deficiencies at billion scale.
- DiskANN (Vamana) [36]: The standard disk-based graph index in Faiss. Since MCGI builds upon the Vamana architecture, this comparison isolates the specific gains derived from our manifold-aware routing strategy.
- IVF-Flat (Faiss) [20]: An in-memory inverted index serving as a performance roofline. Although running in memory-mapped mode, its sequential scanning pattern allows aggressive OS caching, effectively simulating in-memory performance. We include it to benchmark how closely our disk-resident solution approaches the throughput limits of memory-resident systems.

**Evaluation Metrics** We adhere to the standard evaluation protocol for ANN search. We measure Recall@10 against QPS (Queries Per Second). Additionally, we report the query latency at high-recall operating points on the billion-scale datasets.

**Build Configuration** Table 1 summarizes the dataset specifications and indexing parameters used during the construction of MCGI and baseline DiskANN indices. $R$ denotes the maximum outdegree of each node in the graph. $L_{build}$ is the size of the candidate list during index construction. $\alpha$ Range indicates the range of pruning aggressiveness explored during the LID-based Sigmoid mapping. $m_{PQ}$ specifies the number of bytes allocated for Product Quantization (PQ) codes when applicable. Data Type indicates whether the dataset vectors are stored as 32-bit floating-point numbers or 8-bit unsigned integers.

Table 1: Dataset Specifications and Indexing Parameters.

| Dataset | $R$ | $L_{build}$ | $\alpha$ Range | $m_{PQ}$ (Byte) | Data Type |
|---|---|---|---|---|---|
| SIFT1M | {32, 48, 64} | 100 | $[1.0, 1.5]$ | N/A | float32 |
| GloVe-100 | {32, 48, 64} | 100 | $[1.0, 1.5]$ | N/A | float32 |
| SIFT1B | {32, 48} | 50 | $[1.0, 1.5]$ | 16 | uint8 |
| T2I-1B | {32, 48} | 50 | $[1.0, 1.5]$ | 16 | float32 |

**Search Configuration** Table 2 lists the statistical parameters derived from the dataset's LID distribution used in the Sigmoid mapping function $\Phi(\cdot)$, alongside the runtime search sweep parameters. $\mu_{LID}$ and $\sigma_{LID}$ denote the mean and standard deviation of the Local Intrinsic Dimensionality (LID) estimates across the dataset. Beam ($L_{search}$) indicates the range of candidate list sizes evaluated during query execution to capture the Recall-QPS trade-off. Threads specifies the number of parallel threads utilized during search to maximize hardware concurrency.

Table 2: Runtime Search Parameters and LID Statistics.

| Dataset | $\mu_{LID}$ (Mean) | $\sigma_{LID}$ (Std) | Beam ($L_{search}$) | Threads |
|---|---|---|---|---|
| SIFT1M | 14.2 | 3.1 | $10 \rightarrow 200$ | 96 |
| GloVe-100 | 18.5 | 4.2 | $10 \rightarrow 200$ | 96 |
| SIFT1B | 19.5 | 7.9 | $10 \rightarrow 200$ | 96 |
| T2I-1B | 18.3 | 7.0 | $10 \rightarrow 200$ | 96 |

### A.3 Algorithm Pseudo-Code

Algorithm 1 outlines the offline construction process of the indexing structure. The procedure is divided into two sequential phases.

- The first phase focuses on geometric calibration by calculating the Local Intrinsic Dimensionality for all vectors in the dataset in parallel. After obtaining the global mean and standard deviation of these estimates, the algorithm assigns a personalized pruning parameter $\alpha_u$ to each node. This parameter is derived using a logistic function that maps the standardized geometric complexity of the local neighborhood into a bounded operational range.
- The second phase handles topology refinement. Starting from a random graph, the algorithm iteratively optimizes the connections. For each node, it executes a greedy search to identify a pool of candidate neighbors. It then evaluates these candidates alongside existing neighbors using a dynamic occlusion criterion. A candidate is pruned if a previously accepted neighbor falls within a restricted exclusion region defined by the precomputed $\alpha_u$. Survivors are added to the updated neighbor list until the degree limit is reached.

Algorithm 2 presents an optimized online variant designed to handle massive datasets where exhaustive geometric profiling is computationally prohibitive.

- It begins with a fast approximation phase that samples a small subset of the data. By evaluating this subset, the algorithm quickly bootstraps the global mean and standard deviation of the underlying probability distribution. This step provides a coarse but efficient estimate of the overall geometric complexity, enabling the algorithm to initialize the pruning parameters without incurring the cost of a full dataset pass.

**Algorithm 1** Manifold-Consistent Graph Indexing (MCGI)

Input: Dataset $X$, Max Degree $R$, Beam Width $L$
Output: Optimized Graph $G$

// Phase 1: Geometric Calibration
$\mathcal{L} \leftarrow$ ParallelEstimateLID$(X)$
$\mu \leftarrow$ Mean$(\mathcal{L})$
$\sigma \leftarrow$ StdDev$(\mathcal{L})$
**for** each node $u \in V$ in parallel **do**
  $z_u \leftarrow (\mathcal{L}[u] - \mu)/\sigma$
  $\alpha_u \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})/(1 + \exp(z_u))$
**end for**

// Phase 2: Topology Refinement
$G \leftarrow$ RandomGraph$(X, R)$
**for** $iter \leftarrow 1$ to $MaxIter$ **do**
  **for** each node $u \in G$ in parallel **do**
    $\mathcal{C} \leftarrow$ GreedySearch$(u, G, L)$
    $\mathcal{N}_{new} \leftarrow \emptyset$
    **for** $v \in$ SortByDistance$(\mathcal{C} \cup \mathcal{N}(u))$ **do**
      $pruned \leftarrow$ False
      **for** $n \in \mathcal{N}_{new}$ **do**
        **if** $\alpha_u \cdot d(n, v) \leq d(u, v)$ **then**
          $pruned \leftarrow$ True; break
        **end if**
      **end for**
      **if** not $pruned$ **and** $|\mathcal{N}_{new}| < R$ **then**
        $\mathcal{N}_{new}$.add$(v)$
      **end if**
    **end for**
    $\mathcal{N}(u) \leftarrow \mathcal{N}_{new}$
  **end for**
**end for**

- The second phase merges the geometric estimation directly into the iterative topology refinement loop. Instead of computing all complexities beforehand, the algorithm dynamically estimates the node specific dimensionality on the fly using the candidate set discovered during the greedy search step. This real time estimate is immediately used to calculate the specific pruning parameter $\alpha_u$. The algorithm then applies this dynamic parameter to the pruning criterion, determining which edges to preserve and which to discard. This approach ensures the graph connectivity adapts to the manifold geometry progressively as the quality of the candidate neighbors improves with each iteration.

**Algorithm 2** Online Manifold-Consistent Graph Indexing

Input: Dataset $X$, Max Degree $R$, Beam Width $L$, Sample Rate $S$
Output: Optimized Graph $G$

// Phase 1: Bootstrap Statistics (Fast Approximation)
$X_{sample} \leftarrow$ RandomSample$(X, S)$
$\mathcal{L}_{sample} \leftarrow$ EstimateLID$(X_{sample})$ {Compute LID on subset}
$\mu \leftarrow$ Mean$(\mathcal{L}_{sample})$, $\sigma \leftarrow$ StdDev$(\mathcal{L}_{sample})$

// Phase 2: Online Topology Refinement
$G \leftarrow$ RandomGraph$(X, R)$
**for** $iter \leftarrow 1$ to $MaxIter$ **do**
  **for** each node $u \in G$ in parallel **do**
    $\mathcal{C} \leftarrow$ GreedySearch$(u, G, L)$
    // Online LID Estimation & Parameter Adaptation
    $\hat{\ell}_u \leftarrow$ ComputeLID$(u, \mathcal{C})$ {Estimate from neighbors}
    $z_u \leftarrow (\hat{\ell}_u - \mu)/\sigma$
    $\alpha_u \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})/(1 + \exp(z_u))$
    $\mathcal{N}_{new} \leftarrow \emptyset$
    **for** $v \in$ SortByDistance$(\mathcal{C} \cup \mathcal{N}(u))$ **do**
      $pruned \leftarrow$ False
      **for** $n \in \mathcal{N}_{new}$ **do**
        // Pruning with dynamic $\alpha_u$
        **if** $\alpha_u \cdot d(n, v) \leq d(u, v)$ **then**
          $pruned \leftarrow$ True; break
        **end if**
      **end for**
      **if** not $pruned$ **and** $|\mathcal{N}_{new}| < R$ **then**
        $\mathcal{N}_{new}$.add$(v)$
      **end if**
    **end for**
    $\mathcal{N}(u) \leftarrow \mathcal{N}_{new}$
  **end for**
**end for**